Princess Sumaya University for Technology
Communication Engineering Department
Applied Electromagnetics
Worksheet 8

1. Find the force $\boldsymbol{F}$ on a rectangular loop a*b carrying a current I_2 placed parallel to an infinitely long wire carrying a current I_1 at a distant ρ_0 from the near side? Find $\boldsymbol{F}$ if $I_1 = 10\ A$, $I_2 = 5\ A$, $a = 10\ cm$, $b = 30\ cm$ and $\rho_0 = 20\ cm$?

2. Determine the self-inductance of a coaxial cable of inner radius a and radius b?

3. A toroid of radius $\rho_0 = 10\ cm$ and a circular cross section $a = 1\ cm$ is made of steel ($\mu_r = 1000$), and a coil of 200 turns. Calculate the current I required to produce a flux of 0.5 Wb in the core.